

Counting and Combinatorics

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Introduction

This lesson develops the basic toolkit of finite combinatorics: permutations, combinations, multinomial coefficients, the pigeonhole principle, and inclusion–exclusion. The centerpiece is the binomial theorem, which we state and prove by induction. Combinatorial reasoning underlies all of discrete probability and many counting arguments in machine learning theory (sample-space sizes, VC growth functions, counting of hypothesis classes).

First, an example

Before any formulas, let us count by hand. How many ways can we arrange the three letters $\{a, b, c\}$ in a row? List them:

abc, acb, bac, bca, cab, cba.

Read aloud: a-b-c, a-c-b, b-a-c, b-c-a, c-a-b, c-b-a. There are exactly 6 arrangements. Notice the structure: there are 3 choices for the first letter; once chosen, 2 remain for the second; and the last is forced. So $3 \cdot 2 \cdot 1 = 6$. This product pattern is the seed of the multiplication principle, of $n!$, and of the formulas $P(n, k)$ and $\binom{n}{k}$ developed below.

Basic Principles

Definition 1 (Multiplication Principle). If a procedure consists of k stages, with n_i choices available at stage i independently of previous choices, then the total number of ways to execute the procedure is $\prod_{i=1}^k n_i$.

Definition 2 (Addition Principle). If a set S is the disjoint union $S = \bigsqcup_{i=1}^k S_i$, then $|S| = \sum_{i=1}^k |S_i|$.

Permutations and Combinations

Definition 3 (Permutation). Let n, k be nonnegative integers with $k \leq n$. A k -permutation of an n -element set S is an injection $\{1, \dots, k\} \rightarrow S$, equivalently an ordered tuple of k distinct elements of S . The number of k -permutations is

$$P(n, k) = \frac{n!}{(n-k)!} = n(n-1) \cdots (n-k+1).$$

Read aloud: P of n comma k equals n factorial divided by the quantity n minus k factorial, which equals n times n minus 1, dot dot dot, down to n minus k plus 1.

Definition 4 (Combination). A k -combination of an n -element set S is a k -element subset of S . The number of k -combinations is the binomial coefficient

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Read aloud: n choose k equals n factorial divided by the product k factorial times n minus k factorial.

Proposition 1 (Pascal's rule). For $1 \leq k \leq n-1$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Read aloud: n choose k equals n minus 1 choose k minus 1, plus n minus 1 choose k .

Proof. Fix an element $x \in S$ with $|S| = n$. A k -subset either contains x (there are $\binom{n-1}{k-1}$ such, choosing the remaining $k-1$ from $S \setminus \{x\}$) or excludes x (there are $\binom{n-1}{k}$). These cases are disjoint, so the count is the sum. $\square$

The Pigeonhole Principle

Definition 5 (Pigeonhole Principle). If $f: X \rightarrow Y$ is a function between finite sets with $|X| > |Y|$, then f is not injective: there exist $x_1 \neq x_2$ in X with $f(x_1) = f(x_2)$. More generally, if $|X| > k \cdot |Y|$, then some $y \in Y$ has at least $k+1$ preimages.

The Binomial Theorem

Theorem 2 (Binomial Theorem). *For all $n \in \mathbb{Z}_{\geq 0}$ and all x, y in a commutative ring,*

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Read aloud: the quantity x plus y , raised to the n , equals the sum from k equals 0 to n of n choose k , times x to the k , times y to the n minus k .

Proof. We proceed by induction on n .

Base case ($n = 0$): Both sides equal 1. Indeed $(x + y)^0 = 1$ and $\sum_{k=0}^0 \binom{0}{k} x^k y^{-k}$ has its single term $\binom{0}{0} x^0 y^0 = 1$.

Inductive step: Assume the formula holds for some $n \geq 0$. Then

$$\begin{aligned} (x + y)^{n+1} &= (x + y)(x + y)^n = (x + y) \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} \\ &= \sum_{k=0}^n \binom{n}{k} x^{k+1} y^{n-k} + \sum_{k=0}^n \binom{n}{k} x^k y^{n-k+1}. \end{aligned}$$

Read aloud: x plus y to the n plus 1 equals x plus y times x plus y to the n , which equals x plus y times the sum from k equals 0 to n of n choose k , x to the k , y to the n minus k ; distributing gives the sum from k equals 0 to n of n choose k , x to the k plus 1, y to the n minus k , plus the sum from k equals 0 to n of n choose k , x to the k , y to the n minus k plus 1. Reindex the first sum with $j = k + 1$:

$$\sum_{k=0}^n \binom{n}{k} x^{k+1} y^{n-k} = \sum_{j=1}^{n+1} \binom{n}{j-1} x^j y^{n+1-j}.$$

Read aloud: the sum from k equals 0 to n of n choose k , x to the k plus 1, y to the n minus k , equals the sum from j equals 1 to n plus 1 of n choose j minus 1, x to the j , y to the n plus 1 minus j . The second sum is already indexed in the form $\sum_{j=0}^n \binom{n}{j} x^j y^{n+1-j}$. Combining and isolating the $j = 0$ and $j = n + 1$ boundary terms,

$$\begin{aligned} (x + y)^{n+1} &= \binom{n}{n} x^{n+1} y^0 + \sum_{j=1}^n \left[\binom{n}{j-1} + \binom{n}{j} \right] x^j y^{n+1-j} + \binom{n}{0} x^0 y^{n+1} \\ &= x^{n+1} + \sum_{j=1}^n \binom{n+1}{j} x^j y^{n+1-j} + y^{n+1} \end{aligned}$$

Read aloud: x plus y to the n plus 1 equals n choose n times x to the n plus 1 times y to the 0, plus the sum from j equals 1 to n of the quantity n choose j minus 1 plus n choose j , times x to the j , y to the n plus 1 minus j , plus n choose 0 times x to the 0 times y to the n plus 1; this simplifies to x to the n plus 1, plus the sum from j equals 1 to n of n plus 1 choose j , x to the j , y to the n plus 1 minus j , plus y to the n plus 1. by Pascal's rule. Since $\binom{n+1}{0} = \binom{n+1}{n+1} = 1$, this is exactly $\sum_{j=0}^{n+1} \binom{n+1}{j} x^j y^{n+1-j}$, completing the induction. $\square$

Inclusion–Exclusion

Theorem 3 (Inclusion–Exclusion). *For finite sets $A_1, \dots, A_n$,*

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{\emptyset \neq S \subseteq [n]} (-1)^{|S|+1} \left| \bigcap_{i \in S} A_i \right|.$$

Read aloud: the size of the union from i equals 1 to n of A sub i , equals the sum over nonempty subsets S of the index set bracket n , of negative one to the size of S plus 1, times the size of the intersection over i in S of A sub i .

Worked Examples

(a) Permutation. The number of ways to arrange 3 chosen books in order out of 7 available is $P(7, 3) = 7 \cdot 6 \cdot 5 = 210$.

(b) Combination. The number of 5-card poker hands from a standard deck is $\binom{52}{5} = 2,598,960$.

(c) Pigeonhole. Among any 367 people, at least two share a birthday: $|X| = 367 > 366 = |Y|$.

(d) Inclusion–Exclusion. Integers in $\{1, \dots, 100\}$ divisible by 2, 3, or 5:

$$50 + 33 + 20 - 16 - 10 - 6 + 3 = 74.$$

Read aloud: fifty plus thirty-three plus twenty, minus sixteen, minus ten, minus six, plus three, equals seventy-four.

(e) Binomial theorem. Setting $x = y = 1$ in Theorem 2 yields $2^n = \sum_{k=0}^n \binom{n}{k}$, the count of all subsets of an n -set.